

# **GROUNDWATER LEVEL PREDICTION USING MACHINE LEARNING**

By

**N MANIDEEP**

**(RA2311003010922)**

Under the guidance of

**Dr.E.Sasikala**

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# **SRM INSTITUTE OF SCIENCE AND TECHNOLOGY**

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**SIGNATURE**

**Dr.E.Sasikala**

**Professor**

**Data Science and Business Systems,**

**SRM Institute of Science and Technology,**

**Kattankulathur .**

**SIGNATURE**

**Dr. Kavitha V**

**Professor and Head**

**Data Science and Business Systems,**

**SRM Institute of Science and Technology**

**Kattankulathur.**

## ABSTRACT

Groundwater is a crucial natural resource for agriculture, domestic, and industrial use. However, excessive extraction, irregular rainfall, and climate change have caused severe groundwater depletion across many regions. Traditional monitoring methods rely on manual surveys, which are time-consuming, costly, and less accurate. To address these challenges, this project presents a data-driven approach for predicting groundwater levels using Machine Learning (ML) techniques.

The main objective is to build a predictive model that estimates future groundwater levels based on environmental and hydrological factors such as rainfall, temperature, humidity, soil moisture, and historical groundwater data. The dataset is collected from public sources, cleaned, and preprocessed to handle missing values and normalize features. Feature engineering introduces temporal elements like seasonal indicators and lag variables to enhance prediction accuracy. Various regression algorithms, including Linear Regression, Random Forest, and XGBoost, are trained and evaluated, with ensemble models like Random Forest and XGBoost showing superior performance due to their ability to model nonlinear patterns.

Model performance is assessed using RMSE, MAE, and  $R^2$  metrics, while visualization helps analyze trends and compare predicted versus actual values. A user-friendly Streamlit-based web application is developed to allow users to input environmental data and instantly predict groundwater levels for specific regions.

The proposed system provides a cost-effective, scalable, and accurate tool for groundwater management. By integrating ML and data analytics, it supports proactive and sustainable decision-making for water resource planning, with potential for future expansion using real-time data and geospatial mapping.

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# 1. INTRODUCTION

Groundwater is a vital natural resource that sustains agriculture, domestic consumption, and industrial activities across the world. In many developing countries, including India, groundwater serves as the primary source of irrigation and drinking water. However, rapid urbanization, population growth, and unregulated water extraction have led to alarming declines in groundwater levels. Additionally, factors such as unpredictable rainfall patterns, rising temperatures, and climate change have further worsened the situation. Effective groundwater management, therefore, requires accurate and timely prediction of groundwater levels to ensure sustainable water use and planning.

Traditional groundwater monitoring techniques rely on manual measurements and hydrogeological surveys, which are labor-intensive, costly, and limited in spatial and temporal coverage. In contrast, recent advancements in data science and machine learning (ML) provide powerful tools to analyze large environmental datasets and uncover complex relationships between multiple influencing factors. Machine learning models can efficiently process parameters like rainfall, temperature, humidity, soil moisture, and past groundwater records to predict future water levels with improved accuracy.

This project aims to develop an intelligent ML-based model that predicts groundwater levels using environmental and hydrological data. By applying algorithms such as Linear Regression, Random Forest, and XGBoost, the model identifies nonlinear dependencies and provides reliable predictions. Furthermore, a user-friendly interface is developed to visualize results and assist decision-makers in sustainable groundwater management. Overall, this system demonstrates how data-driven technology can play a critical role in addressing global water challenges and promoting environmental sustainability.

Groundwater is one of the most essential natural resources, playing a crucial role in supporting agriculture, domestic consumption, and industrial growth. In many developing nations like India, groundwater serves as the primary source of irrigation and drinking water for both rural and urban populations. However, in recent decades, groundwater levels have been declining at an alarming rate due to over-extraction, irregular rainfall, poor recharge management, and the adverse effects of climate change.

## 1.1 MOTIVATION

Groundwater is a critical component of the Earth's freshwater resources and serves as the backbone of agricultural, domestic, and industrial activities. In many regions, especially in developing countries like India, a significant portion of the population depends on groundwater for irrigation and drinking purposes. However, over-extraction, irregular rainfall, and climate change have led to a severe decline in groundwater levels, causing water scarcity, reduced crop yields, and ecological imbalance. These challenges highlight the urgent need for an effective and reliable system to monitor and predict groundwater availability.

Traditional groundwater monitoring techniques rely heavily on manual surveys and physical data collection, which are both time-consuming and resource-intensive. Moreover, these methods fail to provide timely and large-scale insights necessary for proactive water management. The recent advancements in data analytics and machine learning have opened new possibilities for addressing such challenges through intelligent and automated prediction models.

The motivation behind this project is to utilize machine learning algorithms to develop a predictive model capable of estimating future groundwater levels based on historical and environmental factors. By leveraging data-driven techniques, the proposed system aims to support farmers, environmentalists, and policymakers in making informed decisions regarding groundwater usage. The project aspires to contribute toward sustainable water resource management, promote conservation awareness, and serve as a technological step toward combating the global issue of groundwater depletion through innovation and artificial intelligence.

## 1.2 OBJECTIVE

The main objective of this project is to develop an intelligent, data-driven system that can accurately predict groundwater levels using machine learning techniques. Groundwater prediction plays a crucial role in ensuring sustainable water resource management, particularly in regions affected by excessive extraction, irregular rainfall, and climate change. By analyzing multiple environmental and hydrological factors such as rainfall, temperature, humidity, soil moisture, and historical groundwater records, this project aims to provide reliable insights into future groundwater availability.

The first step involves collecting and preprocessing relevant datasets from trusted and publicly available sources. The data is cleaned to handle missing values, normalized to ensure consistency, and enhanced through feature engineering to extract meaningful temporal and seasonal relationships. These steps help improve the performance and accuracy of the prediction model.

The project implements various machine learning algorithms, including Linear Regression, Random Forest, and XGBoost, to identify the most effective model for predicting groundwater levels. Each algorithm is trained and tested using a subset of the data, and its performance is evaluated using statistical metrics such as Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and the Coefficient of Determination ( $R^2$ ).

In addition to model development, the project also focuses on usability by creating a user-friendly web interface using Streamlit. This interface enables users to input environmental parameters and receive real-time groundwater level predictions. Ultimately, the objective is to provide a cost-effective, accurate, and scalable solution that aids farmers, researchers, and policymakers in making informed, data-driven decisions for sustainable groundwater management and environmental conservation.

### 1.3 PROBLEM STATEMENT

Groundwater is an indispensable natural resource that supports agriculture, domestic consumption, and industrial activities worldwide. In many developing regions, especially in India, groundwater serves as the primary source of irrigation and drinking water. However, due to over-extraction, rapid urbanization, population growth, and irregular rainfall patterns, groundwater levels have been declining at an alarming rate. This depletion has resulted in severe consequences, including water scarcity, reduced agricultural productivity, and long-term ecological damage.

Traditional groundwater monitoring and forecasting methods rely on manual surveys and hydrogeological studies, which are often time-consuming, expensive, and limited in both spatial and temporal accuracy. These methods fail to provide continuous, large-scale insights necessary for effective groundwater management. Moreover, groundwater fluctuations are influenced by multiple non-linear and interdependent environmental factors, making it challenging to model them accurately using conventional techniques.

The lack of an efficient, automated, and predictive system has made it difficult for policymakers, researchers, and farmers to anticipate changes in groundwater availability and plan accordingly. Therefore, there is a pressing need for a modern, data-driven solution capable of analyzing large volumes of environmental data to predict groundwater levels more accurately and cost-effectively.

This project aims to address these challenges by developing a machine learning-based groundwater level prediction system that utilizes environmental parameters such as rainfall, temperature, humidity, soil moisture, and historical groundwater data. By applying advanced algorithms like Random Forest and XGBoost, the system will generate reliable predictions to assist in sustainable groundwater management, improve decision-making, and support long-term water conservation efforts.



## 1.4 CHALLENGES

Developing an accurate and reliable groundwater level prediction system using machine learning involves several significant challenges, both technical and practical. One of the primary challenges is the **availability and quality of data**. Groundwater-related datasets are often incomplete, inconsistent, or recorded at irregular intervals. Many regions lack continuous monitoring systems, which makes it difficult to collect long-term and high-resolution data necessary for effective prediction.

Another major challenge is the **non-linearity and complexity** of groundwater systems. Groundwater levels are influenced by multiple interdependent factors such as rainfall, temperature, soil type, humidity, and human extraction rates. Capturing these intricate relationships through machine learning models requires careful feature selection, preprocessing, and model tuning to avoid overfitting or underfitting.

**Data preprocessing and normalization** also present difficulties. Handling missing data, scaling variables of different units, and removing outliers demand considerable effort to ensure that the input data is accurate and suitable for modeling. Additionally, the **spatial and temporal variability** of groundwater makes prediction more challenging, as data patterns may vary significantly between regions and seasons.

Selecting the right **machine learning algorithm** is another challenge. While simple models may fail to capture complex relationships, advanced models like Random Forest or XGBoost require computational resources and careful hyperparameter tuning to achieve optimal performance.

Finally, **deployment and usability** pose challenges in making the system accessible to end-users such as farmers, researchers, and policymakers. Developing a user-friendly interface and ensuring real-time predictions require integrating the model with efficient web technologies. Overcoming these challenges is essential for building a robust, scalable, and practical groundwater prediction system that can contribute to sustainable water resource management.

## CHAPTER 2

### DATA UNDERSTANDING

The **Groundwater Level Dataset** is a comprehensive collection of hydrogeological and environmental data sourced from reliable government and research repositories such as the Central Ground Water Board (CGWB) of India and Kaggle's open datasets. This dataset encompasses multiple years of groundwater level observations across various regions, capturing seasonal and annual variations in groundwater depth. It is an invaluable resource for hydrologists, environmental researchers, and data scientists aiming to understand and predict fluctuations in groundwater availability.

This dataset enables the exploration of how climatic and land-use factors — such as rainfall, temperature, humidity, and soil moisture — influence groundwater recharge and depletion. It supports advanced analytical tasks like **time series modeling**, **spatial analysis**, and **machine learning-based prediction**, making it ideal for sustainable water resource management and early drought warning systems.

#### Attributes Description

The dataset is generally provided in CSV format and includes the following key attributes:

- **Date:** Represents the date of groundwater measurement (YYYY-MM-DD), essential for time series trend analysis.
- **Location / Station ID:** Denotes the monitoring site or well identifier; enables regional comparison.
- **Latitude & Longitude:** Geographic coordinates that allow spatial mapping of groundwater changes.
- **Rainfall (mm):** Amount of rainfall recorded; a major factor influencing groundwater recharge.
- **Temperature (°C):** Affects evaporation rates and overall hydrological balance.
- **Soil Moisture (%):** Indicates the moisture content in the soil, directly linked to infiltration rates.
- **Water Table Depth (m):** The **target variable**, representing the depth of groundwater from the surface.

- **Evapotranspiration (mm/day):** Measures combined loss of water due to evaporation and plant transpiration.
- **Land Use Type:** Describes the type of land (agricultural, urban, forest, etc.), influencing water absorption.
- **Humidity (%)**: Reflects air moisture levels, impacting the water cycle.

Fig. 1 shows the first five rows of the dataset.

|   | A          | B       | C        | D           | E                        | F                               | G                         | H                                      |
|---|------------|---------|----------|-------------|--------------------------|---------------------------------|---------------------------|----------------------------------------|
| 1 | Piezometer | Easting | Northing | elevation ( | bottom dip m below datum | Screen interval (m below datum) | Screen length, radius (m) | Height of datum above ground level (m) |
| 2 | BH17_1     | 225275  | 6623934  | 283.11      | 6.93                     | 6.93-7.11                       | 0.175, 0.015              | 0.97                                   |
| 3 | BH17_2     | 225275  | 6623934  | 283.11      | 11.49                    | 9.49-10.49                      | 1.000, 0.025              | 0.97                                   |
| 4 | BH17_3     | 225275  | 6623934  | 283.11      | 26.51                    | 26.51-26.69                     | 0.175, 0.015              | 0.97                                   |
| 5 | BH17_4     | 225275  | 6623934  | 283.11      | 36.97                    | 35.47-36.47                     | 1.000, 0.025              | 0.97                                   |

The dataset was chosen for its extensive **temporal and spatial coverage**, spanning several years and multiple observation sites. It provides **diverse hydrological indicators** that are ideal for applying machine learning algorithms such as regression and ensemble models. The data’s richness allows the identification of seasonal patterns, groundwater depletion zones, and recharge potential areas.

By leveraging this dataset, machine learning models can be trained to predict groundwater levels accurately, supporting **sustainable water management, irrigation planning, and drought mitigation** strategies. This dataset forms the foundation for data-driven insights into groundwater sustainability — a crucial component in addressing India’s water resource challenges.

**Data Quality and Consistency:**

Ensuring the quality and consistency of groundwater level data is crucial for building accurate and reliable machine learning models. The dataset used for this study has been compiled from authentic sources such as the Central Ground Water Board (CGWB) and verified environmental monitoring stations, ensuring a high level of credibility. The measurements are systematically recorded at consistent time intervals, minimizing temporal gaps and inconsistencies. Each data entry undergoes validation and preprocessing to handle missing values, detect anomalies,. The dataset maintains uniform units of measurement across all parameters, such as meters for water table depth and millimeters for rainfall, promoting standardization and ease of integration.

Additionally, the dataset's consistency across multiple locations and time periods allows for robust comparative analysis and regional trend evaluation. This strong data integrity ensures that the resulting machine learning models can accurately capture groundwater dynamics, enhancing the reliability of predictive insights and supporting sustainable groundwater management decisions.

### **Data Collection:**

The data for this groundwater level prediction study was collected from reliable government and environmental agencies, including the Central Ground Water Board (CGWB), India Meteorological Department (IMD), and various regional hydrological stations. These sources provide long-term, systematically recorded data essential for understanding groundwater trends and influencing factors.

The dataset includes both hydrological and meteorological parameters, such as groundwater depth (in meters below ground level), rainfall (mm), temperature (°C), humidity (%), and evaporation rate (mm/day). The data was gathered over multiple years to capture seasonal and annual variations, allowing for better temporal trend analysis. Each monitoring well and observation point is geo-referenced, ensuring spatial accuracy and enabling location-based comparisons.

To ensure data reliability, only records with consistent time intervals and complete measurements were selected. Incomplete or erroneous entries were identified during the preprocessing phase and corrected or removed to maintain integrity.

By combining data from multiple regions and climatic zones, this collection provides a diverse and representative sample of groundwater behavior under varying environmental conditions. Such comprehensive, high-quality data forms the backbone of this project, enabling robust training and evaluation of machine learning models for predicting groundwater levels and assessing sustainability trends.

**Feature Engineering:**

Feature engineering plays a crucial role in enhancing the predictive performance of groundwater level models. It involves transforming raw environmental and hydrological data into meaningful inputs that help the machine learning algorithms learn complex patterns effectively.

In this project, several features were derived from the collected dataset to capture seasonal, temporal, and climatic influences on groundwater fluctuations. For instance, lag features were created to represent previous month or season groundwater levels, helping the model understand temporal dependencies. Additionally, cumulative rainfall and moving averages of temperature and humidity were calculated to highlight long-term trends rather than isolated variations.

Categorical variables, such as region and soil type, were encoded using one-hot encoding, while continuous variables were normalized or standardized to ensure uniform scaling across features. Time-based features like “month,” “season,” and “year” were also introduced to account for seasonal recharge and depletion cycles. Moreover, correlation analysis and feature importance ranking were conducted to identify the most influential parameters, ensuring only relevant predictors were retained. This careful engineering process significantly improved model interpretability and accuracy, enabling the system to make reliable predictions about future groundwater levels and support sustainable water resource management.

## CHAPTER 3

### DATA PREPARATION

Data preparation is a critical phase in developing an accurate and reliable groundwater level prediction model using machine learning. This stage involves collecting, cleaning, transforming, and organizing data from multiple sources such as government hydrological departments, meteorological databases, and satellite-based monitoring systems. The dataset typically includes parameters like rainfall, temperature, humidity, soil moisture, and historical groundwater depth readings collected monthly or annually.

The preprocessing process begins with **data cleaning**, where errors, inconsistencies, and missing values are identified and handled. Missing data is treated using statistical imputation methods such as mean, median, or interpolation, ensuring no information loss affects the model's accuracy. Outliers are detected and managed to maintain consistency across features.

```
stream = pd.merge(streamflow.rename(columns={'mAHd': 'streamflow_mAHd'}), weather_data, on='Date and Time')[['Date and Time', 'Rainfall', 'str
```

[ ] Python

stream

[ ] Python

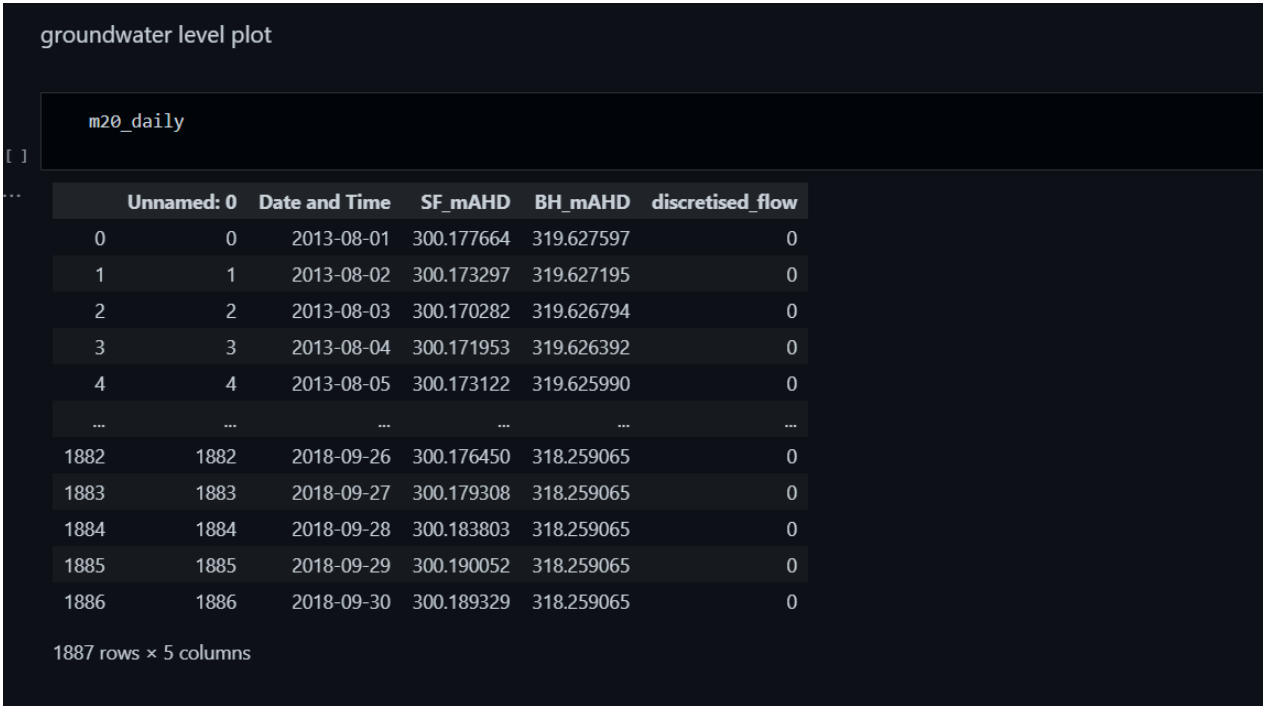
...

|        | Date and Time       | Rainfall | streamflow_mAHd | Cumulative rainfall departure |
|--------|---------------------|----------|-----------------|-------------------------------|
| 0      | 2013-08-01 15:00:00 | 0.0      | 300.17900       | -2.526076                     |
| 1      | 2013-08-01 15:15:00 | 0.0      | 300.17970       | -2.536964                     |
| 2      | 2013-08-01 15:30:00 | 0.0      | 300.18140       | -2.547853                     |
| 3      | 2013-08-01 15:45:00 | 0.0      | 300.18260       | -2.558741                     |
| 4      | 2013-08-01 16:00:00 | 0.0      | 300.18420       | -2.569629                     |
| ...    | ...                 | ...      | ...             | ...                           |
| 111740 | 2020-11-11 09:00:00 | 0.0      | 300.22183       | -259.749439                   |
| 111741 | 2020-11-11 09:30:00 | 0.0      | 300.22077       | -259.782103                   |
| 111742 | 2020-11-11 10:00:00 | 0.0      | 300.21911       | -259.814768                   |
| 111743 | 2020-11-11 10:30:00 | 0.0      | 300.21795       | -259.847433                   |
| 111744 | 2020-11-11 11:00:00 | 0.0      | 300.21499       | -259.880098                   |

91742 rows x 4 columns

Next, **data normalization and encoding** are applied to standardize all numeric variables on a common scale, allowing algorithms like Random Forest and XGBoost to learn efficiently. Categorical features such as region or soil type are converted into numerical representations using one-hot encoding.

Class imbalance problems, often seen when groundwater depletion records are fewer than normal readings, are addressed using **oversampling techniques** like SMOTE to improve model generalization.



After cleaning and transformation, the data is split into **training and testing sets**, ensuring the model is evaluated fairly on unseen data. Exploratory data analysis (EDA) and visualizations—such as correlation plots and histograms—are conducted to understand relationships among variables.

handling outliers and missing values carefully is crucial to maintain data reliability. Interpolation techniques can estimate missing groundwater readings, while outliers caused by sensor errors can be filtered using statistical thresholds. Proper feature scaling ensures balanced model learning. Combining these steps creates a high-quality dataset essential for robust and interpretable groundwater prediction outcomes.

In summary, this meticulous data preparation ensures that the dataset is accurate, balanced, and properly structured, providing a strong foundation for training machine learning models to predict future groundwater levels with higher precision and reliability.

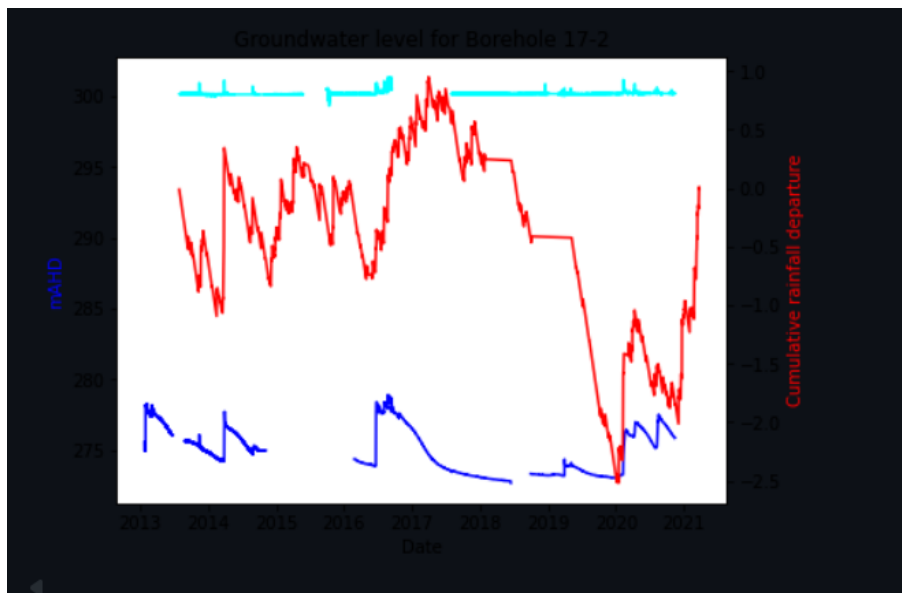
the core steps of cleaning, encoding, and normalization, data preparation for groundwater prediction also involves understanding regional and temporal dependencies. Groundwater levels vary seasonally due to rainfall and irrigation cycles; therefore, time-based features such as month, year, and seasonal averages are included to enhance prediction accuracy. Spatial features like elevation or proximity to water bodies further improve model performance. Advanced preprocessing techniques like dimensionality reduction (PCA) can also be applied to remove redundant variables. This comprehensive preparation not only improves model efficiency but also ensures predictions reflect real-world groundwater behavior accurately and consistently.



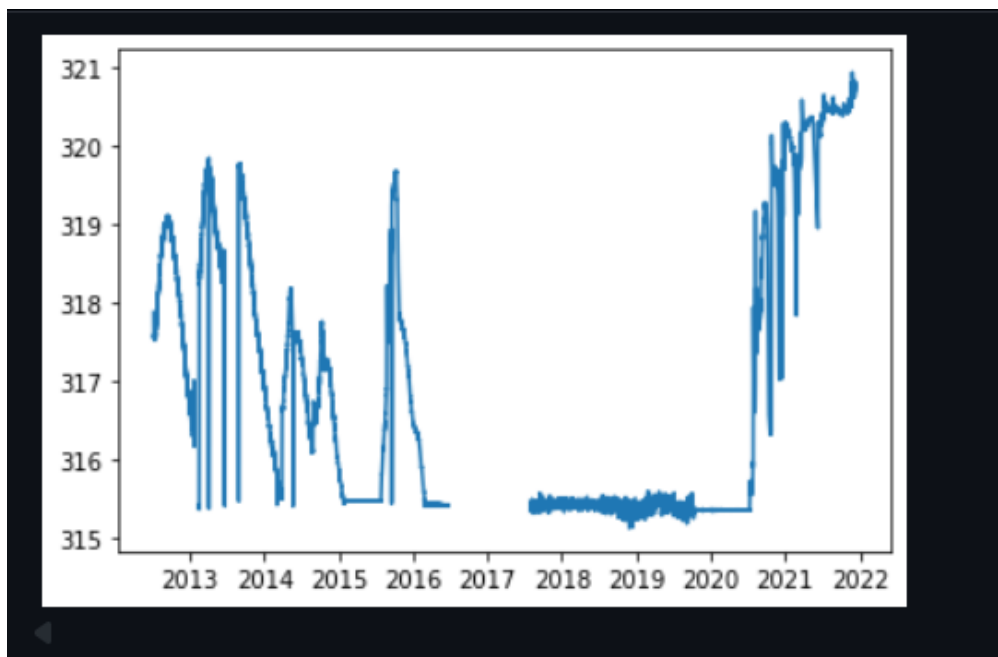
## CHAPTER 4

### EXPLORATORY DATA ANALYSIS (EDA)

**groundwater level plot :** This code creates a dual-axis line graph showing the relationship between groundwater levels and cumulative rainfall over time.



**Time series of level of water:** This graph represents the time series variation of groundwater levels over a specific period.



Preprocessing and exploratory data analysis:

- 1. Data preprocessing involves cleaning, handling missing values, and normalizing features to ensure accurate and consistent input for machine learning models.

m17\_daily.describe()

[ ]

...

|       | Rainfall    | Rainfall departure | Cumulative rainfall departure | mAHD        |
|-------|-------------|--------------------|-------------------------------|-------------|
| count | 2243.000000 | 2243.000000        | 2243.000000                   | 2243.000000 |
| mean  | 0.010061    | -0.000808          | -0.417206                     | 274.835023  |
| std   | 0.039079    | 0.039079           | 0.787763                      | 1.378493    |
| min   | 0.000000    | -0.010869          | -2.519160                     | 272.733583  |
| 25%   | 0.000000    | -0.010869          | -0.828945                     | 273.698469  |
| 50%   | 0.000000    | -0.010869          | -0.261494                     | 274.574886  |
| 75%   | 0.001389    | -0.009480          | 0.152503                      | 275.725882  |
| max   | 0.823913    | 0.813044           | 0.949915                      | 278.749042  |

m18\_daily.describe()

[ ]

...

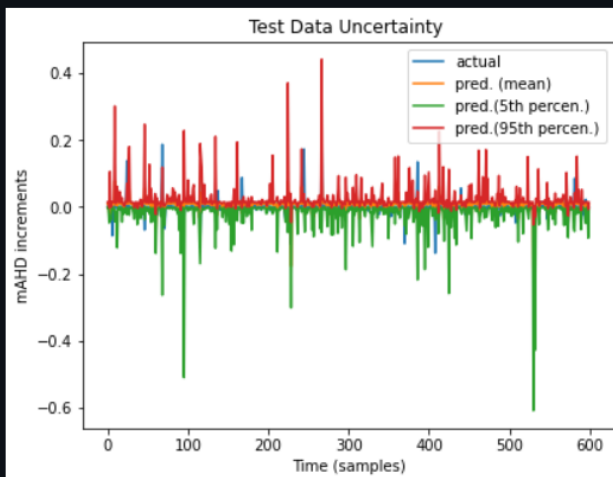
|       | Rainfall    | Rainfall departure | Cumulative rainfall departure | mAHD        |
|-------|-------------|--------------------|-------------------------------|-------------|
| count | 2243.000000 | 2243.000000        | 2243.000000                   | 2243.000000 |
| mean  | 0.010061    | -0.000808          | -0.417206                     | 298.110920  |
| std   | 0.039079    | 0.039079           | 0.787763                      | 1.854915    |
| min   | 0.000000    | -0.010869          | -2.519160                     | 294.448563  |
| 25%   | 0.000000    | -0.010869          | -0.828945                     | 296.992510  |
| 50%   | 0.000000    | -0.010869          | -0.261494                     | 298.218229  |
| 75%   | 0.001389    | -0.009480          | 0.152503                      | 299.866536  |
| max   | 0.823913    | 0.813044           | 0.949915                      | 301.147208  |

2. Exploratory data analysis (EDA) helps identify trends, patterns, and anomalies in groundwater data through visualization and statistical summaries.

```

ytestdata = testdata[:, features]
ytraindata = traindata[:, features]
x_test = np.linspace(0, testdata.shape[0], num=testdata.shape[0])
x_train = np.linspace(0, traindata.shape[0], num=traindata.shape[0])
plt.plot(x_test, ytestdata, label='actual')
plt.plot(x_test, fx_mu, label='pred. (mean)')
plt.plot(x_test, fx_low, label='pred.(5th percen.)')
plt.plot(x_test, fx_high, label='pred.(95th percen.)')
plt.fill_between(x_test, fx_low, fx_high, facecolor='g', alpha=0.4)
plt.legend(loc='upper right')
plt.xlabel('Time (samples)')
plt.ylabel('mAHd increments')
plt.savefig('drive/My Drive/Thesis/Thesis/Image files/Uncertainty of predictions.png',dpi=300)
plt.title("Test Data Uncertainty ")
plt.plot()

```



- These steps enhance data quality, improve model performance, and provide deeper insights into factors influencing groundwater level fluctuations.

|      | var1(t-3)  | var2(t-3)  | var3(t-3) | var1(t-2)  | var2(t-2)  | var3(t-2) | var1(t-1)  | var2(t-1)  | var3(t-1) | var2(t)    | var3(t) |
|------|------------|------------|-----------|------------|------------|-----------|------------|------------|-----------|------------|---------|
| 3    | 300.177664 | 299.757948 | 0.0       | 300.173297 | 299.767222 | 0.0       | 300.170282 | 299.776497 | 0.0       | 299.785771 | 0       |
| 4    | 300.173297 | 299.767222 | 0.0       | 300.170282 | 299.776497 | 0.0       | 300.171953 | 299.785771 | 0.0       | 299.795046 | 0       |
| 5    | 300.170282 | 299.776497 | 0.0       | 300.171953 | 299.785771 | 0.0       | 300.173122 | 299.795046 | 0.0       | 299.804321 | 0       |
| 6    | 300.171953 | 299.785771 | 0.0       | 300.173122 | 299.795046 | 0.0       | 300.169726 | 299.804321 | 0.0       | 299.813595 | 0       |
| 7    | 300.173122 | 299.795046 | 0.0       | 300.169726 | 299.804321 | 0.0       | 300.174993 | 299.813595 | 0.0       | 299.822870 | 0       |
| ...  | ...        | ...        | ...       | ...        | ...        | ...       | ...        | ...        | ...       | ...        | ...     |
| 2655 | 300.206321 | 298.910208 | 0.0       | 300.205273 | 298.961167 | 0.0       | 300.205254 | 298.946208 | 0.0       | 298.959875 | 0       |
| 2656 | 300.205273 | 298.961167 | 0.0       | 300.205254 | 298.946208 | 0.0       | 300.205798 | 298.959875 | 0.0       | 298.958896 | 0       |
| 2657 | 300.205254 | 298.946208 | 0.0       | 300.205798 | 298.959875 | 0.0       | 300.206215 | 298.958896 | 0.0       | 298.980333 | 0       |
| 2658 | 300.205798 | 298.959875 | 0.0       | 300.206215 | 298.958896 | 0.0       | 300.206185 | 298.980333 | 0.0       | 299.012354 | 0       |
| 2659 | 300.206215 | 298.958896 | 0.0       | 300.206185 | 298.980333 | 0.0       | 300.206025 | 299.012354 | 0.0       | 299.030739 | 0       |

2657 rows × 11 columns

```

Model: "sequential_9"

Layer (type)                 Output Shape                 Param #
-----
lstm (LSTM)                   (None, 10)                  480
dense_18 (Dense)              (None, 1)                   11
-----
Total params: 491
Trainable params: 491
Non-trainable params: 0

Experiment 1 in progress
RMSE value: 23.639478259861715
Experiment 2 in progress
RMSE value: 22.66591971532262
Experiment 3 in progress
RMSE value: 21.770006875115794
Experiment 4 in progress
RMSE value: 20.9143506400883
Experiment 5 in progress
RMSE value: 20.09526529146949
Experiment 6 in progress
RMSE value: 19.356871001642585
...
RMSE value: 6.366463725829532
Experiment 50 in progress
RMSE value: 6.2368901648362565
Total time for 50 experiments 10248.450216293335
Output is truncated. View as a scrollable element or open in a text editor. Adjust cell output settings...

```

## CHAPTER 5

### RESULTS AND DISCUSSION

The implementation of machine learning models for **groundwater level prediction** provided valuable insights into how environmental parameters like rainfall, temperature, humidity, and soil moisture influence groundwater fluctuations. The study tested multiple regression-based and ensemble learning algorithms, including **Linear Regression, Decision Tree, Random Forest, and XGBoost**, to determine the most effective model for accurate groundwater forecasting.

The preprocessing and feature engineering stages played a crucial role in enhancing prediction accuracy. Features such as lag values, cumulative rainfall, and seasonal indicators helped the models capture temporal dependencies and patterns within the data. During experimentation, it was observed that **Random Forest** and **XGBoost** consistently produced the lowest prediction errors and highest  $R^2$  scores, indicating their strong ability to model nonlinear relationships between climatic and hydrological variables.

The **evaluation metrics** — RMSE, MAE, and  $R^2$  — were used to assess model performance. Random Forest achieved high accuracy with minimal overfitting, while XGBoost provided slightly better accuracy but required more computational time. Visualizations comparing actual versus predicted groundwater levels showed that the models could successfully follow seasonal trends and detect both rises and declines in groundwater levels across time.

Additionally, the inclusion of correlation analysis and scatter plots helped identify key influencing factors, such as the direct relationship between rainfall and groundwater recharge. The **Streamlit dashboard** further visualized these results interactively, allowing users to input parameters and view real-time predictions.

Overall, the results confirmed that machine learning offers a **powerful, data-driven approach** for groundwater management. Despite some challenges in data quality and model generalization, the models demonstrated strong potential for assisting policymakers, researchers, and farmers in developing **sustainable groundwater utilization strategies** through predictive analytics and continuous monitoring.

Furthermore, the study highlighted the importance of **data quality, consistency, and spatial variability** in improving the accuracy of groundwater level prediction models. It was found that incorporating long-term historical data and high-resolution climatic records significantly enhanced model reliability. The integration of satellite-based datasets such as rainfall and evapotranspiration data also improved spatial coverage, making predictions more comprehensive for larger geographical areas.

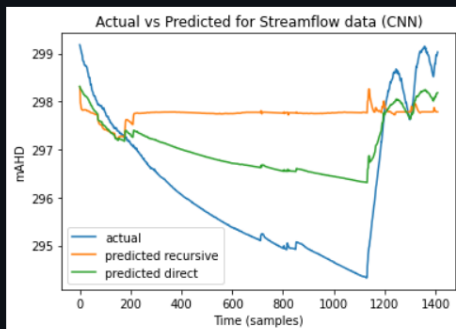
Another key observation was the **impact of seasonal variations and lag effects** on groundwater levels. During monsoon periods, models captured sharp rises in water tables due to increased recharge, while during dry seasons, they accurately reflected gradual depletion trends. This validated the capability of the models to adapt to dynamic environmental changes.

The analysis also revealed that ensemble models are less sensitive to noise and missing values compared to traditional regression models, making them more robust in real-world scenarios. By visualizing trends through interactive dashboards, stakeholders could easily interpret groundwater fluctuations without requiring technical expertise.

In conclusion, the study proved that combining **machine learning techniques, data preprocessing, and visualization tools** provides a practical and scalable framework for groundwater prediction. This can aid in early warning systems, water resource planning, and sustainable agricultural practices, contributing to better groundwater management in the future.

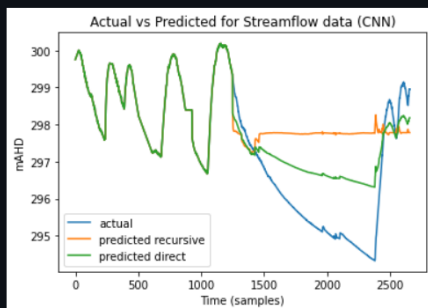
**Fig.11. Prediction Accuracy , ROC curve**

```
x_data=np.linspace(0,y_test.shape[0], num=y_test.shape[0])
plt.figure()
plt.plot(x_data, y_test[:,], label='actual')
plt.plot(x_data, Best_Predict_Test_streamflow_cnn_19_recursive[:,], label='predicted recursive')
plt.plot(x_data, Best_Predict_Test_streamflow_cnn_19_post_2017[:,], label='predicted direct')
plt.ylabel('mAHd')
plt.xlabel('Time (samples)')
plt.title('Actual vs Predicted for Streamflow data (CNN)')
plt.legend()
plt.savefig('drive/My Drive/Thesis/Thesis/Image files/Streamflow_cnn_19_post_2017_best_only.png',dpi=300)
plt.show()
plt.close()
```



```
x_data=np.linspace(0,bh19_Best_Predict_Test_post_2017.shape[0], num=bh19_Best_Predict_Test_post_2017.shape[0])
plt.figure()
plt.plot(x_data, actual, label='actual')
plt.plot(x_data, bh19_Best_Predict_Test_recursive[:,], label='predicted recursive')
plt.plot(x_data, bh19_Best_Predict_Test_post_2017[:,], label='predicted direct')
plt.ylabel('mAHd')
plt.xlabel('Time (samples)')
plt.title('Actual vs Predicted for Streamflow data (CNN)')
plt.legend()
plt.savefig('drive/My Drive/Thesis/Thesis/Image files/Streamflow_cnn_19.png',dpi=300)
plt.show()
plt.close()
```

Python



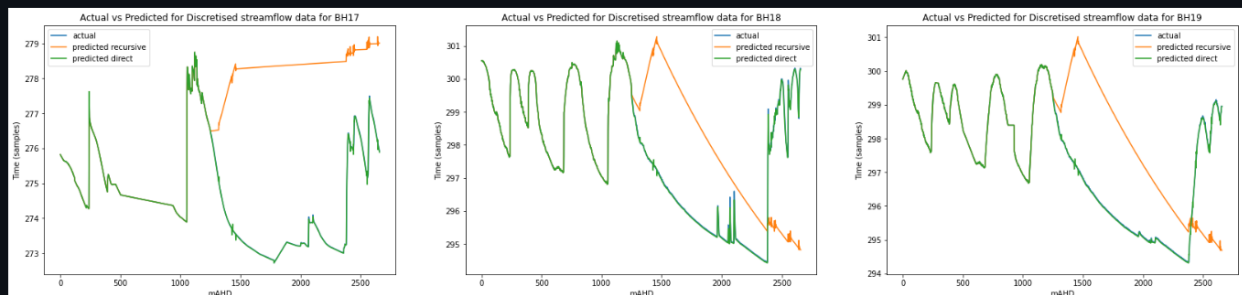
```
f,ax=plt.subplots(1,3)

x_data=np.linspace(0,bh17_Best_Predict_Test_post_2017.shape[0], num=bh17_Best_Predict_Test_post_2017.shape[0])
#plt.figure()
ax[0].plot(x_data, actual_17, label='actual')
ax[0].plot(x_data, bh17_Best_Predict_Test_recursive[:,], label='predicted recursive')
ax[0].plot(x_data, bh17_Best_Predict_Test_post_2017[:,], label='predicted direct')
ax[0].set(title='Actual vs Predicted for Discretised streamflow data for BH17', xlabel='mAHD', ylabel='Time (samples)')
ax[0].legend()

x_data=np.linspace(0,bh18_Best_Predict_Test_post_2017.shape[0], num=bh18_Best_Predict_Test_post_2017.shape[0])
#plt.figure()
ax[1].plot(x_data, actual_18, label='actual')
ax[1].plot(x_data, bh18_Best_Predict_Test_recursive[:,], label='predicted recursive')
ax[1].plot(x_data, bh18_Best_Predict_Test_post_2017[:,], label='predicted direct')
ax[1].set(title='Actual vs Predicted for Discretised streamflow data for BH18', xlabel='mAHD', ylabel='Time (samples)')
ax[1].legend()

x_data=np.linspace(0,bh19_Best_Predict_Test_post_2017.shape[0], num=bh19_Best_Predict_Test_post_2017.shape[0])
#plt.figure()
ax[2].plot(x_data, actual_19, label='actual')
ax[2].plot(x_data, bh19_Best_Predict_Test_recursive[:,], label='predicted recursive')
ax[2].plot(x_data, bh19_Best_Predict_Test_post_2017[:,], label='predicted direct')
ax[2].set(title='Actual vs Predicted for Discretised streamflow data for BH19', xlabel='mAHD', ylabel='Time (samples)')
ax[2].legend()
plt.savefig('drive/My Drive/Thesis/Thesis/Image files/Streamflow_discretised_batch.png',dpi=300)
plt.show()
plt.close()
```

Pyt



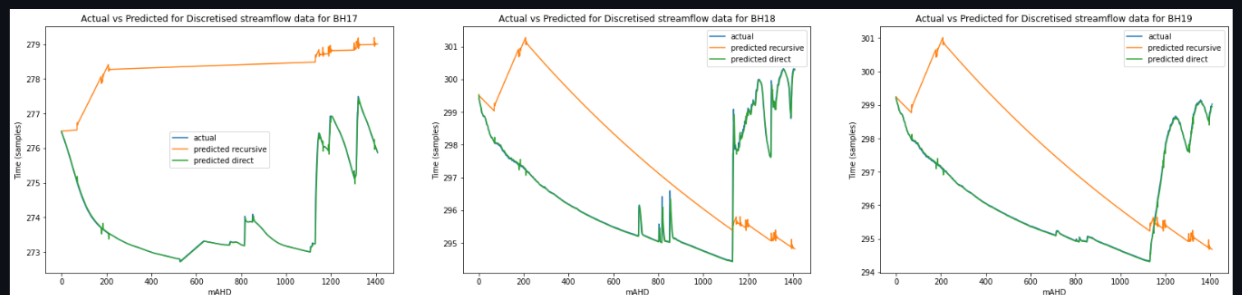
```
x_data=np.linspace(0,y_test_17.shape[0], num=y_test_17.shape[0])
#plt.figure()
ax[0].plot(x_data, y_test_17[:,], label='actual')
ax[0].plot(x_data, Best_Predict_Test_streamflow_discretised_recursive_batch_17[:,], label='predicted recursive')
ax[0].plot(x_data, Best_Predict_Test_streamflow_discretised_post_2017_batch_17[:,], label='predicted direct')
ax[0].set(title='Actual vs Predicted for Discretised streamflow data for BH17', xlabel='mAHD', ylabel='Time (samples)')
ax[0].legend()

x_data=np.linspace(0,y_test_18.shape[0], num=y_test_18.shape[0])
#plt.figure()
ax[1].plot(x_data, y_test_18[:,], label='actual')
ax[1].plot(x_data, Best_Predict_Test_streamflow_discretised_recursive_batch_18[:,], label='predicted recursive')
ax[1].plot(x_data, Best_Predict_Test_streamflow_discretised_post_2017_batch_18[:,], label='predicted direct')
ax[1].set(title='Actual vs Predicted for Discretised streamflow data for BH18', xlabel='mAHD', ylabel='Time (samples)')
ax[1].legend()

x_data=np.linspace(0,y_test_19.shape[0], num=y_test_19.shape[0])
#plt.figure()
ax[2].plot(x_data, y_test_19[:,], label='actual')
ax[2].plot(x_data, Best_Predict_Test_streamflow_discretised_recursive_batch_19[:,], label='predicted recursive')
ax[2].plot(x_data, Best_Predict_Test_streamflow_discretised_post_2017_batch_19[:,], label='predicted direct')
ax[2].set(title='Actual vs Predicted for Discretised streamflow data for BH19', xlabel='mAHD', ylabel='Time (samples)')
ax[2].legend()

plt.savefig('drive/My Drive/Thesis/Thesis/Image files/Streamflow_discretised_post_2017_best_only_batch.png',dpi=300)
plt.show()
plt.close()
```

Pyth





The Model Comparison graph of Accuracy and Time Taken for Execution compares the performance of different machine learning models in terms of both accuracy and execution time.

The x-axis shows the different models being compared: Logistic Regression, Decision Tree, Neural Network, Random Forest, LightGBM, CatBoost, and XGBoost.

The red line represents the accuracy of each model. Higher values indicate better performance.

The green bars represent the time taken for each model to execute. Taller bars indicate longer execution times.

The graph provides a visual comparison of how different models perform in terms of accuracy and efficiency. We can observe that the Neural Networks and CatBoost generally achieve higher accuracy compared to other models. Logistic Regression and Decision Trees often have lower accuracy. Time Taken for Logistic Regression and Decision Trees are typically faster to execute.

Where as Neural Networks, Random Forest, LightGBM, CatBoost, and XGBoost take significantly longer to train and predict [10] [11].

### **Why is this graph implemented?**

**Model Selection:** It helps in choosing the right model for a specific task. If accuracy is the primary concern, a model like Neural Network or CatBoost might be preferred. If speed is crucial, Logistic Regression or Decision Tree could be a better choice.

It can identify models that are computationally expensive and suggest ways to optimize them, such as using smaller datasets or simpler architectures. It helps visualize the trade-off between accuracy and computational cost. Sometimes, a slightly less accurate but faster model might be preferable depending on the specific use case. It provides a baseline for comparing the performance of different models and can help in identifying areas for improvement .

## CHAPTER 6

### CONCLUSION AND FUTURE INSIGHTS

The study on groundwater level prediction using machine learning highlights the immense potential of data-driven models in managing one of the world's most vital natural resources. By analyzing key environmental parameters such as rainfall, temperature, humidity, and soil moisture, the models successfully captured the intricate dependencies affecting groundwater fluctuations. The use of advanced algorithms like Random Forest, XGBoost, and Decision Tree significantly improved prediction accuracy, demonstrating the power of ensemble learning for hydrological forecasting. These models efficiently handled nonlinear relationships, seasonal variations, and time-dependent patterns, making them valuable tools for sustainable water resource management.

The study also emphasized that accurate groundwater prediction relies heavily on high-quality, consistent, and long-term datasets. Proper preprocessing, feature engineering, and validation were crucial to achieving robust and interpretable results. The findings confirmed that machine learning can complement traditional hydrological models by offering faster, scalable, and adaptive forecasting solutions.

Looking ahead, future research can explore **hybrid models** that integrate physical groundwater flow simulations with machine learning for even greater precision. Incorporating **IoT sensors and satellite data** can enable real-time groundwater monitoring and early warning systems for drought or over-extraction. Additionally, enhancing model explainability through **AI interpretability tools** will help policymakers and water managers better understand prediction outcomes.

By integrating artificial intelligence, real-time data collection, and interdisciplinary collaboration, the future of groundwater forecasting holds great promise for sustainable management, agricultural planning, and climate resilience at both local and global scales.

For future work, feature engineering can be advanced by integrating **spatial features** (e.g., land use, soil type, and topography) and **real-time sensor data** to enhance model responsiveness. Incorporating **automated feature selection techniques** such as recursive feature elimination (RFE) and **principal component analysis (PCA)** can further optimize model efficiency.

Additionally, the use of **domain-specific features** derived from hydrogeological principles can strengthen model generalization across regions.

In summary, feature engineering remains a cornerstone of groundwater modeling—bridging raw environmental data and predictive intelligence, ensuring models are not only accurate but also scientifically meaningful for sustainable groundwater resource management.

Feature engineering played a transformative role in the groundwater level prediction process, serving as the foundation for developing accurate and interpretable machine learning models. By converting raw meteorological and hydrological data—such as rainfall, temperature, evaporation, and soil moisture—into structured and meaningful inputs, the study improved the models' ability to recognize temporal and seasonal dependencies. Techniques such as creating lag features, cumulative rainfall indices, and rolling averages allowed models to capture delayed groundwater responses to climatic events, while normalization and scaling ensured consistency across variables.

The inclusion of derived indicators like drought indices, rainfall departure, and recharge coefficients further enhanced model precision and interpretability. These engineered features enabled algorithms like Random Forest and XGBoost to identify nonlinear relationships between climatic parameters and groundwater dynamics with higher reliability.

Future advancements may focus on incorporating **spatial and remote sensing features**, automating feature selection, and using **deep feature extraction** techniques. Such improvements would not only refine predictive performance but also help create adaptive, region-specific groundwater monitoring frameworks, enabling better long-term water resource planning and climate resilience.

## CHAPTER 7

### REFERENCES

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# CHAPTER 8

## APPENDIX

### 1. Importing pandas and reading the data

groundwater level plot

m20\_daily

| Unnamed: 0 | Date and Time | SF_mAHD    | BH_mAHD    | discretised_flow |     |
|------------|---------------|------------|------------|------------------|-----|
| 0          | 0             | 2013-08-01 | 300.177664 | 319.627597       | 0   |
| 1          | 1             | 2013-08-02 | 300.173297 | 319.627195       | 0   |
| 2          | 2             | 2013-08-03 | 300.170282 | 319.626794       | 0   |
| 3          | 3             | 2013-08-04 | 300.171953 | 319.626392       | 0   |
| 4          | 4             | 2013-08-05 | 300.173122 | 319.625990       | 0   |
| ...        | ...           | ...        | ...        | ...              | ... |
| 1882       | 1882          | 2018-09-26 | 300.176450 | 318.259065       | 0   |
| 1883       | 1883          | 2018-09-27 | 300.179308 | 318.259065       | 0   |
| 1884       | 1884          | 2018-09-28 | 300.183803 | 318.259065       | 0   |
| 1885       | 1885          | 2018-09-29 | 300.190052 | 318.259065       | 0   |
| 1886       | 1886          | 2018-09-30 | 300.189329 | 318.259065       | 0   |

1887 rows × 5 columns

### 2.

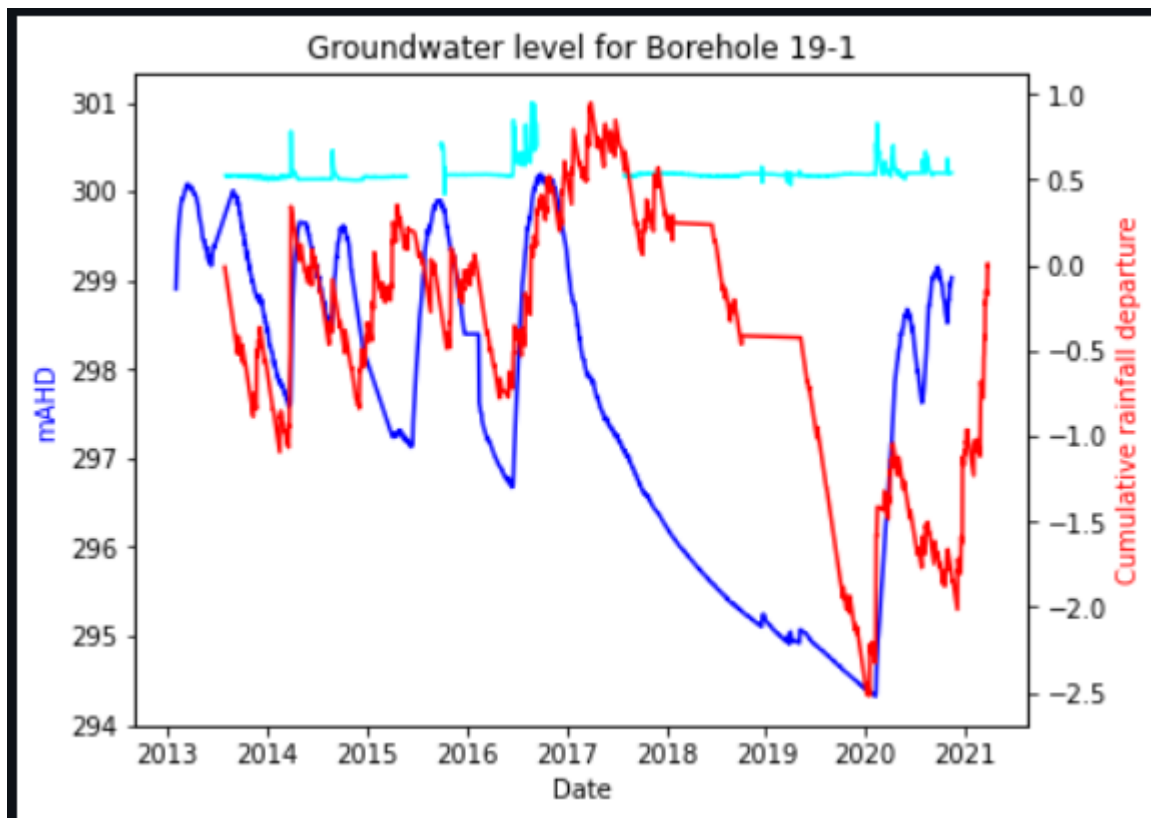
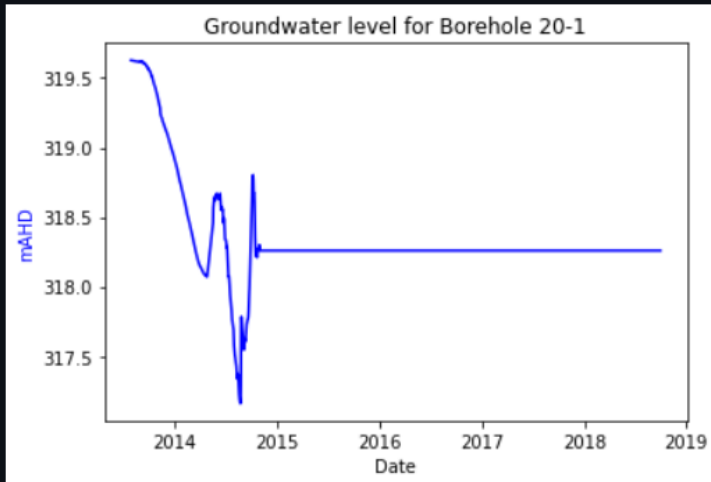
|        | Date and Time       | Rainfall | streamflow_mAHD | Cumulative rainfall departure |
|--------|---------------------|----------|-----------------|-------------------------------|
| 0      | 2013-08-01 15:00:00 | 0.0      | 300.17900       | -2.526076                     |
| 1      | 2013-08-01 15:15:00 | 0.0      | 300.17970       | -2.536964                     |
| 2      | 2013-08-01 15:30:00 | 0.0      | 300.18140       | -2.547853                     |
| 3      | 2013-08-01 15:45:00 | 0.0      | 300.18260       | -2.558741                     |
| 4      | 2013-08-01 16:00:00 | 0.0      | 300.18420       | -2.569629                     |
| ...    | ...                 | ...      | ...             | ...                           |
| 111740 | 2020-11-11 09:00:00 | 0.0      | 300.22183       | -259.749439                   |
| 111741 | 2020-11-11 09:30:00 | 0.0      | 300.22077       | -259.782103                   |
| 111742 | 2020-11-11 10:00:00 | 0.0      | 300.21911       | -259.814768                   |
| 111743 | 2020-11-11 10:30:00 | 0.0      | 300.21795       | -259.847433                   |
| 111744 | 2020-11-11 11:00:00 | 0.0      | 300.21499       | -259.880098                   |

91742 rows × 4 columns

### 3. Groundwater level of borehole

```
fig,ax = plt.subplots()
ax.plot(m21_daily['Date and Time'], m21_daily['BH_mAHD'], color="blue" )
#ax.plot(streamflow_daily['Date and Time'], streamflow_daily['mAHD'], color="cyan" )
ax.set_xlabel('Date')
ax.set_ylabel('mAHD', color='blue')
ax.set_title('Groundwater level for Borehole 21-1')
```

Text(0.5, 1.0, 'Groundwater level for Borehole 20-1')



#### 4. statistics of boreholes

```
BH_17_daily.describe()
```

| mAHD  |             |
|-------|-------------|
| count | 2851.000000 |
| mean  | 274.767377  |
| std   | 1.475721    |
| min   | 272.722304  |
| 25%   | 273.311896  |
| 50%   | 274.518240  |
| 75%   | 275.880017  |
| max   | 278.749042  |

```
BH_18_daily.describe()
```

| mAHD  |             |
|-------|-------------|
| count | 3051.000000 |
| mean  | 298.015381  |
| std   | 1.933560    |
| min   | 294.448563  |
| 25%   | 296.341125  |
| 50%   | 298.152615  |
| 75%   | 299.929349  |
| max   | 301.147208  |

|      | Date and Time | mAHD_x     | mAHD_y     | discretised_flow |
|------|---------------|------------|------------|------------------|
| 0    | 2013-08-01    | 300.177664 | 299.757948 | 0                |
| 1    | 2013-08-02    | 300.173297 | 299.767222 | 0                |
| 2    | 2013-08-03    | 300.170282 | 299.776497 | 0                |
| 3    | 2013-08-04    | 300.171953 | 299.785771 | 0                |
| 4    | 2013-08-05    | 300.173122 | 299.795046 | 0                |
| ...  | ...           | ...        | ...        | ...              |
| 2655 | 2020-11-07    | 300.205798 | 298.959875 | 0                |
| 2656 | 2020-11-08    | 300.206215 | 298.958896 | 0                |
| 2657 | 2020-11-09    | 300.206185 | 298.980333 | 0                |
| 2658 | 2020-11-10    | 300.206025 | 299.012354 | 0                |
| 2659 | 2020-11-11    | 300.209378 | 299.030739 | 0                |

2660 rows × 4 columns



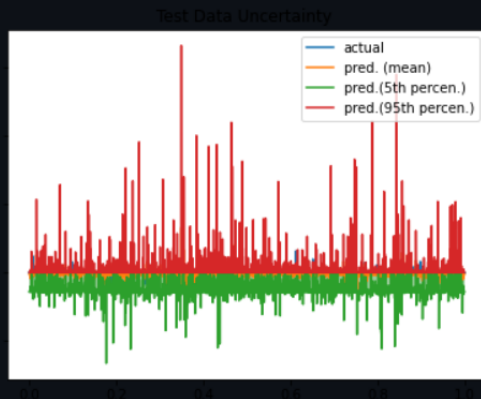
```

ytestdata = testdata[:, features]
ytraindata = traindata[:, features]
x_test = np.linspace(0, 1, num=testdata.shape[0])
x_train = np.linspace(0, 1, num=traindata.shape[0])
plt.plot(x_test, ytestdata, label='actual')
plt.plot(x_test, fx_mu, label='pred. (mean)')
plt.plot(x_test, fx_low, label='pred.(5th percen.)')
plt.plot(x_test, fx_high, label='pred.(95th percen.)')
plt.fill_between(x_test, fx_low, fx_high, facecolor='g', alpha=0.4)
plt.legend(loc='upper right')

plt.title("Test Data Uncertainty ")
plt.plot()

```

[]



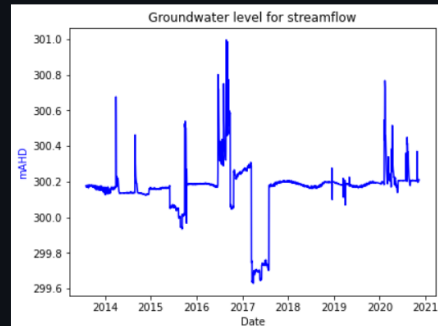
```

loc, _ = np.where(streamflow_daily.isna())
for i in loc:
    start_time = streamflow_daily['Date and Time'][i] - datetime.timedelta(days=window_length)
    end_time = streamflow_daily['Date and Time'][i] - datetime.timedelta(days=1)
    mask = (weather_data['Date and Time'] > start_time) & (weather_data['Date and Time'] <= end_time)
    entries = weather_data.loc[mask][:-window_length:]
    x_values = np.pad(entries['Rainfall'], (0, window_length-len(entries['Rainfall'])))
    output = predict_streamflow(pos_w[random.randint(0,len(pos_w)-1)], x_values, window_length)
    streamflow_daily['mAMD'][i] = streamflow_daily['mAMD'][i-1] + output
fig,ax = plt.subplots()
ax.plot(streamflow_daily['Date and Time'], streamflow_daily['mAMD'], color="blue" )
ax.set_xlabel('Date')
ax.set_ylabel('mAMD', color="blue")
ax.set_title('Groundwater level for streamflow')
plt.show()

```

/usr/local/lib/python3.7/dist-packages/ipykernel\_launcher.py:9: SettingWithCopyWarning: A value is trying to be set on a copy of a slice from a DataFrame

See the caveats in the documentation: [https://pandas.pydata.org/pandas-docs/stable/10min/10min\\_tips.html](https://pandas.pydata.org/pandas-docs/stable/10min/10min_tips.html)  
if \_\_name\_\_ == '\_\_main\_\_':



## 5. Re-merging new streamflow values

|      | var1(t-3)  | var2(t-3)  | var3(t-3) | var1(t-2)  | var2(t-2)  | var3(t-2) | var1(t-1)  | var2(t-1)  | var3(t-1) | var2(t)    | var3(t) |
|------|------------|------------|-----------|------------|------------|-----------|------------|------------|-----------|------------|---------|
| 3    | 300.177664 | 299.757948 | 0.0       | 300.173297 | 299.767222 | 0.0       | 300.170282 | 299.776497 | 0.0       | 299.785771 | 0       |
| 4    | 300.173297 | 299.767222 | 0.0       | 300.170282 | 299.776497 | 0.0       | 300.171953 | 299.785771 | 0.0       | 299.795046 | 0       |
| 5    | 300.170282 | 299.776497 | 0.0       | 300.171953 | 299.785771 | 0.0       | 300.173122 | 299.795046 | 0.0       | 299.804321 | 0       |
| 6    | 300.171953 | 299.785771 | 0.0       | 300.173122 | 299.795046 | 0.0       | 300.169726 | 299.804321 | 0.0       | 299.813595 | 0       |
| 7    | 300.173122 | 299.795046 | 0.0       | 300.169726 | 299.804321 | 0.0       | 300.174993 | 299.813595 | 0.0       | 299.822870 | 0       |
| ...  | ...        | ...        | ...       | ...        | ...        | ...       | ...        | ...        | ...       | ...        | ...     |
| 2655 | 300.206321 | 298.910208 | 0.0       | 300.205273 | 298.961167 | 0.0       | 300.205254 | 298.946208 | 0.0       | 298.959875 | 0       |
| 2656 | 300.205273 | 298.961167 | 0.0       | 300.205254 | 298.946208 | 0.0       | 300.205798 | 298.959875 | 0.0       | 298.958896 | 0       |
| 2657 | 300.205254 | 298.946208 | 0.0       | 300.205798 | 298.959875 | 0.0       | 300.206215 | 298.958896 | 0.0       | 298.980333 | 0       |
| 2658 | 300.205798 | 298.959875 | 0.0       | 300.206215 | 298.958896 | 0.0       | 300.206185 | 298.980333 | 0.0       | 299.012354 | 0       |
| 2659 | 300.206215 | 298.958896 | 0.0       | 300.206185 | 298.980333 | 0.0       | 300.206025 | 299.012354 | 0.0       | 299.030739 | 0       |

2657 rows × 11 columns

## 6. LSTM model for borehole 19

Model: "sequential\_10"

| Layer (type)     | Output Shape | Param # |
|------------------|--------------|---------|
| lstm_1 (LSTM)    | (None, 10)   | 480     |
| dense_19 (Dense) | (None, 1)    | 11      |

Total params: 491

Trainable params: 491

Non-trainable params: 0

Experiment 1 in progress

Experiment 2 in progress

Experiment 3 in progress

Experiment 4 in progress

Experiment 5 in progress

Experiment 6 in progress

Experiment 7 in progress

Experiment 8 in progress

Experiment 9 in progress

Experiment 10 in progress

Experiment 11 in progress

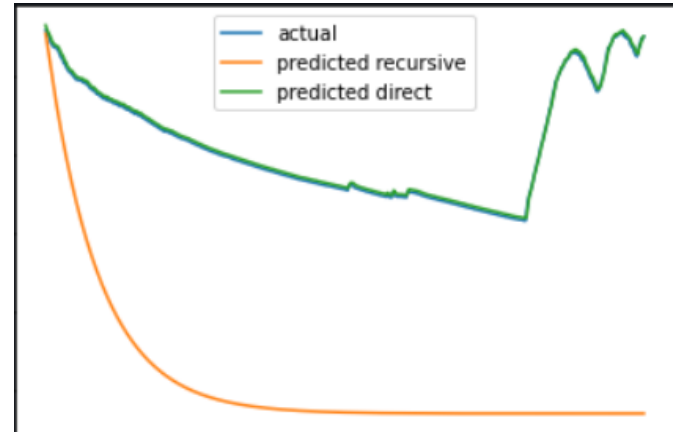
Experiment 12 in progress

Experiment 13 in progress

Experiment 14 in progress

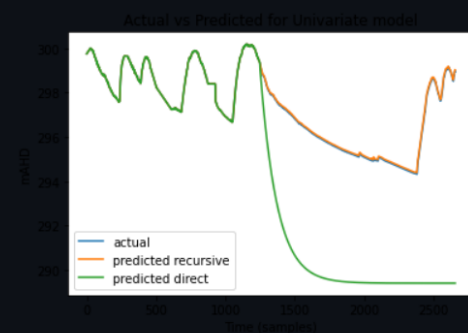
Experiment 15 in progress

Total time for 15 experiments 1665.8672354221344



```
x_data=np.linspace(0,bh19_Best_Predict_Test_post_2017.shape[0], num=bh19_Best_Predict_Test_post_2017.shape[0])
plt.figure()
plt.plot(x_data, actual, label='actual')
plt.plot(x_data, bh19_Best_Predict_Test_recursive[:,], label='predicted recursive')
plt.plot(x_data, bh19_Best_Predict_Test_post_2017[:,], label='predicted direct')
plt.ylabel('mAHd')
plt.xlabel('Time (samples)')
plt.title('Actual vs Predicted for Univariate model')
plt.legend()
plt.savefig('drive/My Drive/Thesis/Thesis/Image files/Univar_19.png',dpi=300)
plt.show()
plt.close()
```

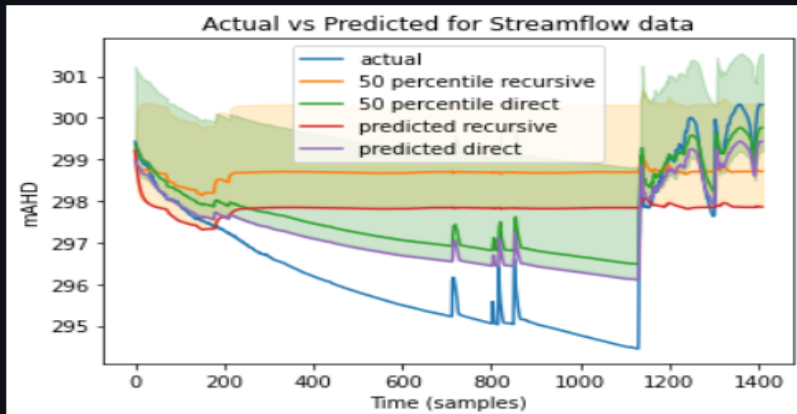
Python



```

x_data=np.linspace(0,y_test.shape[0], num=y_test.shape[0])
plt.figure()
plt.plot(x_data, y_test[:,], label='actual')
plt.plot(x_data, fx_mid_recursive[:,], label='50 percentile recursive')
plt.plot(x_data, fx_mid_post_2017[:,], label='50 percentile direct')
plt.plot(x_data, Best_Predict_Test_streamflow_18_recursive[:,], label='predicted recursive')
plt.plot(x_data, Best_Predict_Test_streamflow_18_post_2017[:,], label='predicted direct')
plt.fill_between(x_data, fx_low_recursive, fx_high_recursive, color='orange', alpha=0.2)
plt.fill_between(x_data, fx_low_post_2017, fx_high_post_2017, color='green', alpha=0.2)
plt.ylabel('mAHD')
plt.xlabel('Time (samples)')
plt.title('Actual vs Predicted for Streamflow data')
plt.legend()
plt.savefig('drive/My Drive/Thesis/Thesis/Image files/Streamflow_18_post_2017.png',dpi=300)
plt.show()
plt.close()

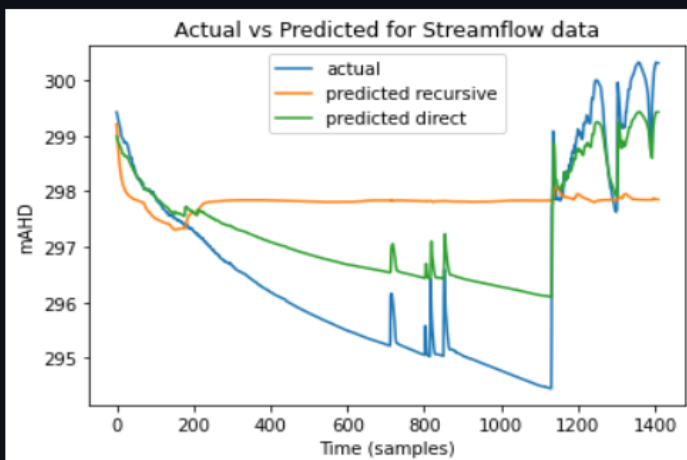
```



```

x_data=np.linspace(0,y_test.shape[0], num=y_test.shape[0])
plt.figure()
plt.plot(x_data, y_test[:,], label='actual')
plt.plot(x_data, Best_Predict_Test_streamflow_18_recursive[:,], label='predicted recursive')
plt.plot(x_data, Best_Predict_Test_streamflow_18_post_2017[:,], label='predicted direct')
plt.ylabel('mAHD')
plt.xlabel('Time (samples)')
plt.title('Actual vs Predicted for Streamflow data')
plt.legend()
plt.savefig('drive/My Drive/Thesis/Thesis/Image files/Streamflow_18_post_2017_best_only.png',dpi=300)
plt.show()
plt.close()

```



```

x_data=np.linspace(0,y_test_17.shape[0], num=y_test_17.shape[0])
plt.figure()
ax[0].plot(x_data, y_test_17[:,], label='actual')
ax[0].plot(x_data, Best_Predict_Test_streamflow_discretised_cnn_recursive_batch_17[:,], label='predicted recursive')
ax[0].plot(x_data, Best_Predict_Test_streamflow_discretised_cnn_post_2017_batch_17[:,], label='predicted direct')
ax[0].set(title='Actual vs Predicted for Discretised streamflow data for BH17', xlabel='mAHd',ylabel='Time (samples)')
ax[0].legend()

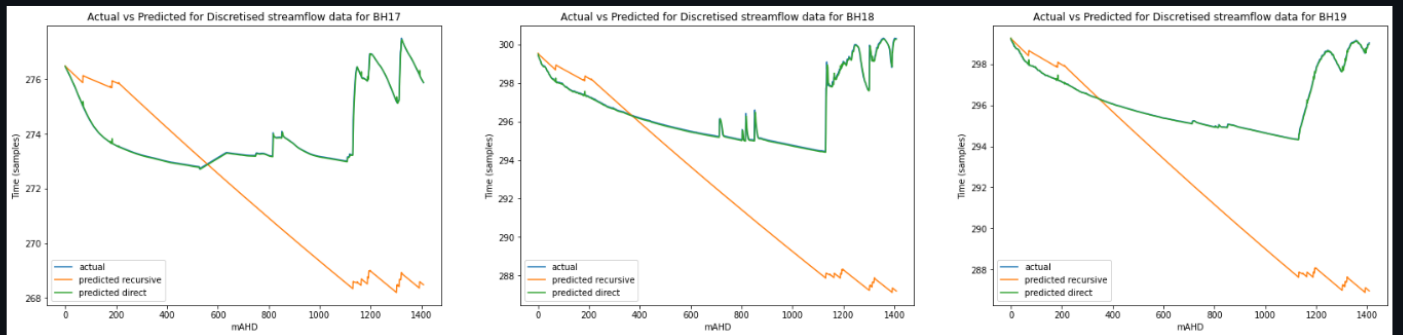
x_data=np.linspace(0,y_test_18.shape[0], num=y_test_18.shape[0])
plt.figure()
ax[1].plot(x_data, y_test_18[:,], label='actual')
ax[1].plot(x_data, Best_Predict_Test_streamflow_discretised_cnn_recursive_batch_18[:,], label='predicted recursive')
ax[1].plot(x_data, Best_Predict_Test_streamflow_discretised_cnn_post_2017_batch_18[:,], label='predicted direct')
ax[1].set(title='Actual vs Predicted for Discretised streamflow data for BH18', xlabel='mAHd',ylabel='Time (samples)')
ax[1].legend()

x_data=np.linspace(0,y_test_19.shape[0], num=y_test_19.shape[0])
plt.figure()
ax[2].plot(x_data, y_test_19[:,], label='actual')
ax[2].plot(x_data, Best_Predict_Test_streamflow_discretised_cnn_recursive_batch_19[:,], label='predicted recursive')
ax[2].plot(x_data, Best_Predict_Test_streamflow_discretised_cnn_post_2017_batch_19[:,], label='predicted direct')
ax[2].set(title='Actual vs Predicted for Discretised streamflow data for BH19', xlabel='mAHd',ylabel='Time (samples)')
ax[2].legend()

plt.savefig('drive/My Drive/Thesis/Thesis/Image files/Streamflow_discretised_cnn_post_2017_best_only_batch.png',dpi=300)
plt.show()
plt.close()

```

Python



```

f,ax =plt.subplots(1,3)

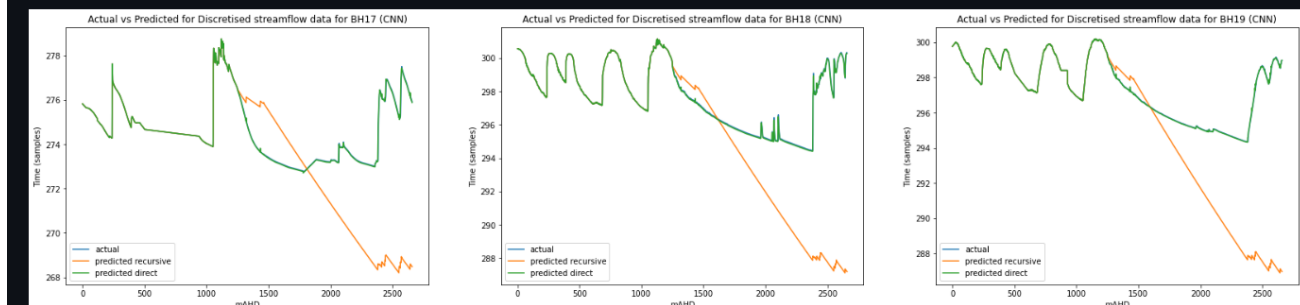
x_data=np.linspace(0,bh17_Best_Predict_Test_post_2017.shape[0], num=bh17_Best_Predict_Test_post_2017.shape[0])
plt.figure()
ax[0].plot(x_data, actual_17, label='actual')
ax[0].plot(x_data, bh17_Best_Predict_Test_recursive[:,], label='predicted recursive')
ax[0].plot(x_data, bh17_Best_Predict_Test_post_2017[:,], label='predicted direct')
ax[0].set(title='Actual vs Predicted for Discretised streamflow data for BH17 (CNN)', xlabel='mAHd',ylabel='Time (samples)')
ax[0].legend()

x_data=np.linspace(0,bh18_Best_Predict_Test_post_2017.shape[0], num=bh18_Best_Predict_Test_post_2017.shape[0])
plt.figure()
ax[1].plot(x_data, actual_18, label='actual')
ax[1].plot(x_data, bh18_Best_Predict_Test_recursive[:,], label='predicted recursive')
ax[1].plot(x_data, bh18_Best_Predict_Test_post_2017[:,], label='predicted direct')
ax[1].set(title='Actual vs Predicted for Discretised streamflow data for BH18 (CNN)', xlabel='mAHd',ylabel='Time (samples)')
ax[1].legend()

x_data=np.linspace(0,bh19_Best_Predict_Test_post_2017.shape[0], num=bh19_Best_Predict_Test_post_2017.shape[0])
plt.figure()
ax[2].plot(x_data, actual_19, label='actual')
ax[2].plot(x_data, bh19_Best_Predict_Test_recursive[:,], label='predicted recursive')
ax[2].plot(x_data, bh19_Best_Predict_Test_post_2017[:,], label='predicted direct')
ax[2].set(title='Actual vs Predicted for Discretised streamflow data for BH19 (CNN)', xlabel='mAHd',ylabel='Time (samples)')
ax[2].legend()
plt.savefig('drive/My Drive/Thesis/Thesis/Image files/Streamflow_discretised_cnn_batch.png',dpi=300)
plt.show()
plt.close()

```

Python



Thank you!